Sequential Decision Making and Explanations

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Outline

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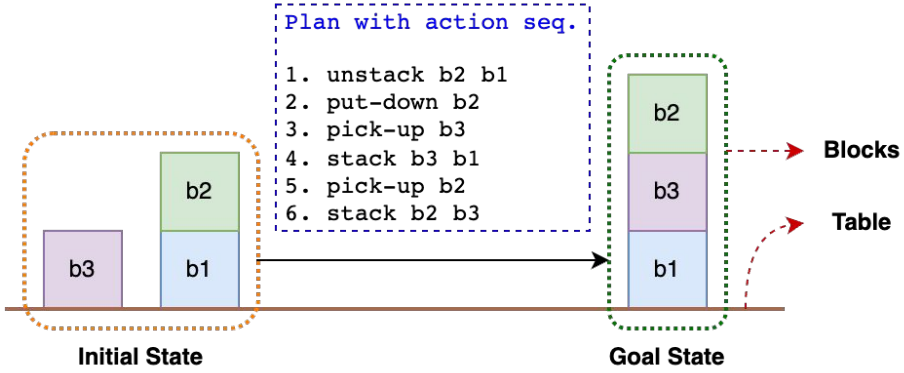
3. Classical Planning
4. PDDL - Example
5. Heuristics
6. Explanation Methodologies
7. Towards Real-World Planning

Background

Automated Planning: also known as AI planning, is the process of finding a sequence of actions that will transform an initial state of the world into a desired goal state, given the world model.

Key elements: Goals, Actions, States, Constraints

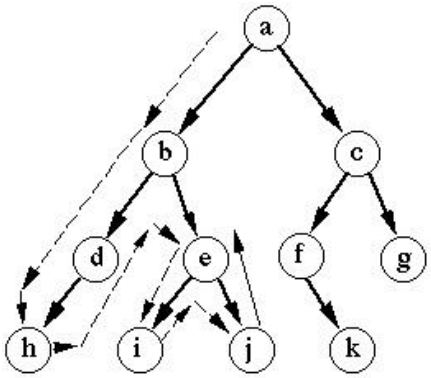
"Selecting a goal-leading course of action based on a high-level description of the world."



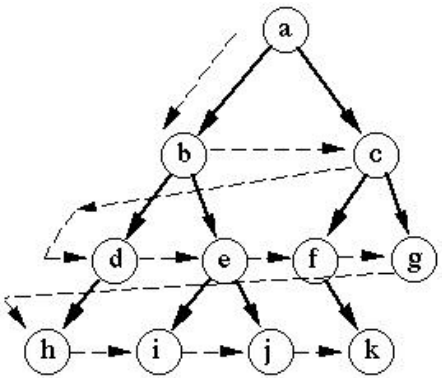
P = (F,O,I,G)

- ➔ **F** : Fluents (state variables)
- ➔ **O** : Operators (actions with preconditions & effects)
- ➔ **I** : Initial state
- ➔ **G** : Goal condition
- ➔ **Planner:** Finds a sequence of actions transforming I → G

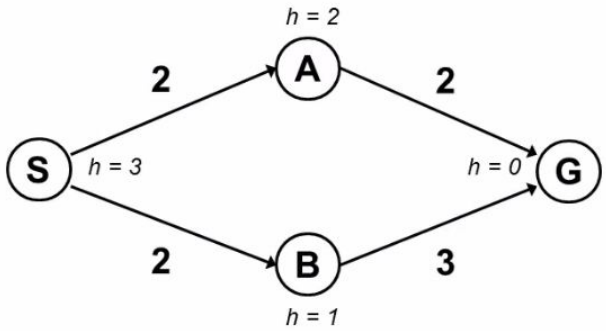
Search



Depth First Search



Breadth First Search



A* Search

Category of Informed search.

f(n) = g(n) + h(n)

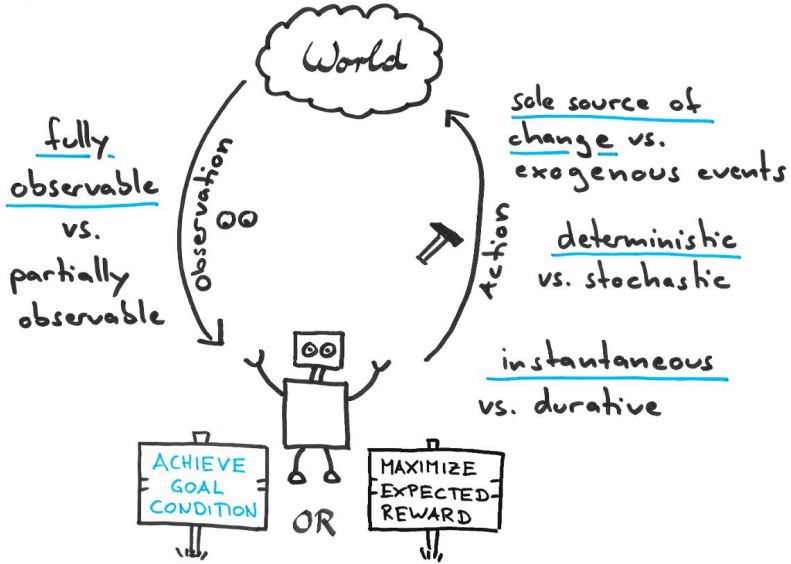
g(n) = Cost-to-node, h(n) = Cost-to-go/ ***Heuristic***

Ref: Google Images

Classical Planning

- **F:** Fluents (state variables) / Predicates
  - A state defines which fluents are true at any given point.
- **O:** Operators (actions with preconditions & effects) / Actions
  - pre(o) ⊆ F → set of fluents that must be true for o to be applicable in state s
  - eff(o) ⊆ F → set of fluents changed in a state (either added or removed) when action is applied.
  - o = [pre(o), eff+(o), eff−(o)]
- **I:** Initial state
  - Represents initial state (set of fluents)
- **G:** Goal State
- **Planner:** Finds a sequence of actions transforming I → G
  - $\pi = (o1, o2, \ldots, on)$
  - $s_0 = I$
  - $s_{k+1} = [s_k \setminus eff^-(o_{k+1}) \cup eff^+(o_{k+1})]$

Classical Planning



Ref: https://ai.dmi.unibas.ch/_files/teaching/hs25/po/slides/po-a02-handout.pdf

PDDL – Example

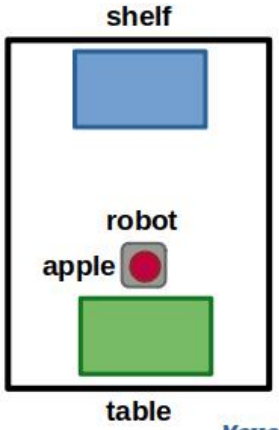
Predicates describe the world state:

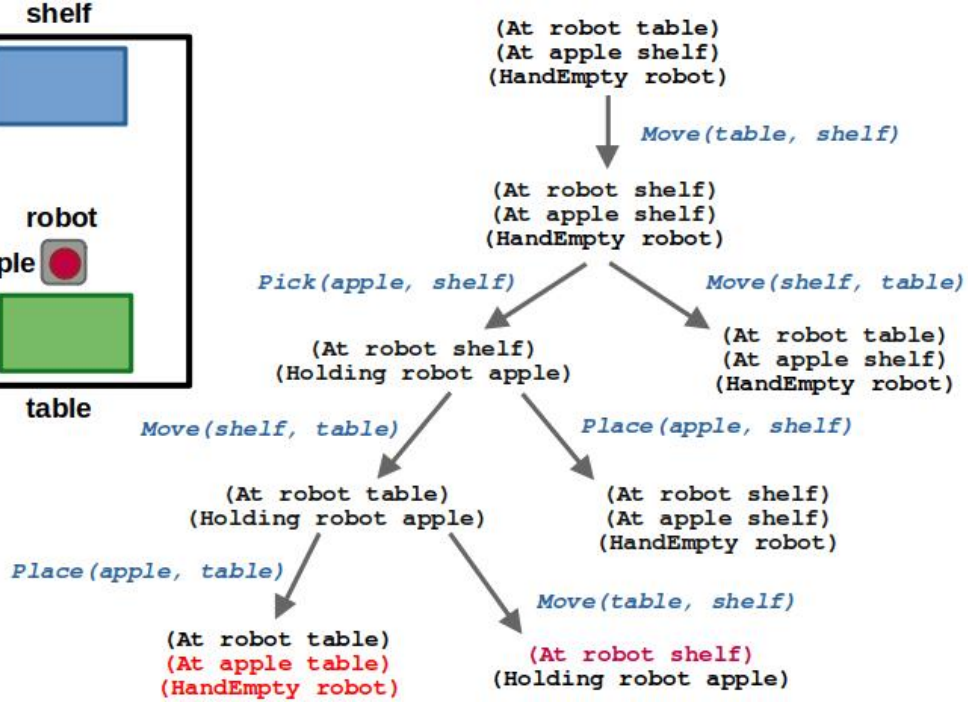
- *(At x l)* → x is at location l
- *(Holding robot o)* → robot holds object o
- *(HandEmpty robot)* → robot is not holding anything

Actions change the state:

- *Move(from, to)* moves the robot between locations
- *Pick(object, location)* lets robot pick an object if co-located and hand is empty
- *Place(object, location)* puts the object down at the current location

Planning searches for a sequence of these actions that transforms the initial state into the goal state.





Ref: Google Images

Heuristics

Heuristic Category	Specific Heuristic(s)	Core Principle	Admissibility	Typical Use Case/Planner
Delete-Relaxation	hadd, hmax, hFF (Relaxed Plan Graph)	Ignore delete effects of actions; solve simplified problem.	hmax: Yes. hadd: No. hFF: No.	FF (hFF), HSP (hadd,hmax), many modern planners.
Landmark	hLM, hLM–Cut	Identify facts that must be true in any solution; count remaining landmarks or use ordering.	Some variants are admissible.	LAMA planner, satisficing and optimal planning.
Abstraction / PDBs	Pattern Databases (PDBs)	Project problem onto a subset of variables (pattern); solve abstract problem optimally; store costs.	Yes (if abstract solution is optimal and costs combined correctly).	Optimal planning, puzzles (e.g., 15-puzzle).
Causal Graph	Causal Graph Heuristic	Based on SAS+ formalism; analyze dependencies between multi-valued variables.	Typically inadmissible but informative.	Fast Downward planner.

Heuristics guide search ($h(s)$). Admissible ($h(s) \leq h^*(s)$)

1. ***Delete-Relaxation:*** hmax (admissible), hadd (informative), hFF (effective) [10].
2. ***Abstraction (PDBs):*** Optimal costs in abstract space.
3. ***Landmarks:*** Facts/actions in every plan.

Explanations - Methodologies

1. **Model-Based Explanations:** These explanations utilize the planning model itself (e.g., action preconditions and effects, causal links).
   a. *Causal Explanations:* Identify the causal links between actions, states, and goals. For example, explaining why an action was necessary to achieve a specific precondition of a subsequent action.
   b. *Contrastive Explanations:* Explain why a particular plan was chosen over another plan, or why a certain action was included/excluded. This often involves highlighting the critical differences that make the chosen plan superior or the other plan infeasible or suboptimal.
   c. *Why-Not Explanations:* Address why a desired state or action is not part of the plan, or why a seemingly viable alternative plan was not generated.
2. **Plan-Based Explanations:** These focus on properties of the generated plan itself.
   a. *Descriptive Explanations:* Summarize the plan's structure, objectives, or key sub-sequences of actions.
   b. *Justification of Action Choices:* Provide reasons for selecting specific actions at particular steps in the plan.
3. **Explanations via Interpretable Models:** This involves designing planning systems that are inherently more transparent.
   a. *Rule-Based Systems:* Plans derived from clear, human-readable rules
   b. *Reinforcement Learning Policy Summarization:* For planners based on RL, techniques to summarize or visualize the learned policy can serve as explanations.

Demo – Single Agent – Explanation Generation

(Single Agent) Planning Ontology Tool - https://editor.planning.domains/#

Additional: Wrappers for Planning

- Frameworks/ Libraries for AI Planning
  - Unified Planning - A python library for AI Planning - https://unified-planning.readthedocs.io/en/latest/getting_started.html
  - Plan Utils

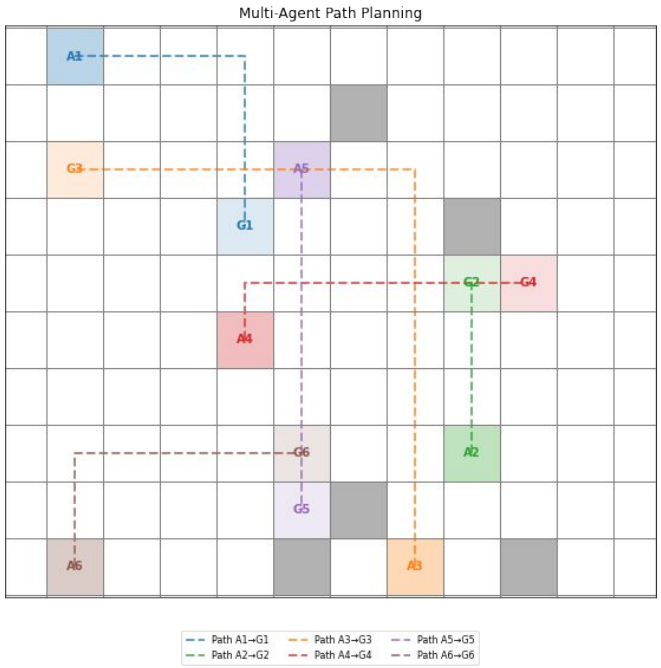
Problem

Definition: Multi-Agent Path Finding (MAPF) asks how to move multiple agents from their start locations to their goal locations on a shared graph or grid without collisions.

Constraints:

1. Agents typically occupy cells in discrete time steps.
2. No two agents should occupy the same cell at the same time.

Goal: Minimize some objective (e.g., the total number of time steps, makespan, or sum of individual path costs).



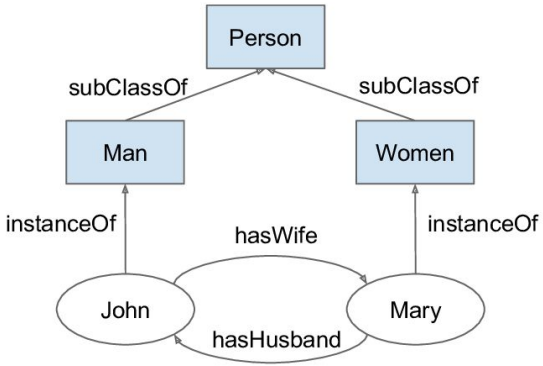
Demo - Multi-Agent - Explanation Generation

- Multi-agent Planning
- Plan Explanations for Multi-Agent Path Finding (MAPF) - https://ai4society.github.io/ma-planning-ontology/#demo

Background - Ontologies

Ontologies: are structured and formal representations of knowledge that capture concepts, relationships, and constraints within a specific domain.

Key elements: Concepts, Relationships, Attributes, Constraints.



Challenges

Real-world environments feature:

1. *Unpredictable Events*
2. *Non-Deterministic Actions*
3. *Partial Observability*
4. *Continuous Changes*
5. *Moving Obstacles*
6. *Imperfect Execution*

These make classical plans fail. *"Dynamics"* encompasses external changes, action uncertainty, and incomplete knowledge.

Planning in Dynamic Scenarios

1. **Core Strategies for Dynamic Planning**
   a. *Replanning:* Re-invoke the planner from the current state upon failure [24].
   b. *Plan Repair/Adaptation:* Make limited modifications to the existing plan [24]. Cheaper, less disruptive.
   c. *Dynamic Replanning Schedulers:* Combine offline planning with online response (rerouting, error handling)
2. **Policy-Based and Proactive Planning Approaches**
   a. *Policy-Based Planning:* Output is a policy $\pi(s \rightarrow a)$ [2]. Common in MDPs/POMDPs [2] and RL [33].
   b. *Proactive Planning:* Creates robust offline plans anticipating disruptions [32]. Involves foresight, risk mitigation [32].
3. **Representing Dynamic and Hybrid Systems: PDDL+**
   a. PDDL+ [5] extends PDDL for hybrid systems [5]. It includes continuous effects [5], timed initial literals [5], and exogenous activity: Events (instantaneous changes) [5] and Processes (continuous changes via ODEs) [5]
4. **Temporal and Probabilistic Frameworks**
   a. *Temporal:* action durations and concurrency
   b. *Probabilistic:* MDPs, POMDPs

MACHINE LEARNING TECHNIQUES

ML Application Area	Common ML Techniques/Approaches	Target of Learning	Source of Experience	Key Contributions/Examples (Ref.)
Heuristic Function Learning	Linear Regression, Neural Networks, Decision Trees	Heuristic function ($h(s)$)	Solved problem instances, Relaxed Plans (RPs)	Learning from RP features [13]; PDBs for numeric planning [42].
Search Control Learning	Rule Learning (e.g., Association Rules), Reinforcement Learning (RL), Supervised Classification/Regression	Policies ($\pi(s \rightarrow a)$), Control Rules, Value Functions	Solved instances, Expert traces, Planner execution traces	Adaptive planner parameter tuning [43]; Control for temporally extended goals [44]; NMP [29]; RL for policies [33].
Action Model Learning	Inductive Logic Programming (ILP), Bayesian Learning, Clustering, Supervised Learning for predicate mapping	Action preconditions, effects, probabilities	Observed state transitions, Interaction with environment	OBSERVER, ARMS, TRAIL, LOPE, PELA [12]; Learning visual predicates [45].
Plan Adaptation & Repair	Reinforcement Learning, Case-Based Reasoning (CBR), Supervised Learning for failure prediction	Repair policies, Adaptation rules, Failure predictors	Execution traces, Simulated failures, Expert repairs	RL for adaptive policies [47]; CBR e-learning route adaptation [30]; Predicting failure modes [49].

References

Need to fix

THANK YOU ALL

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